Methods

This prospective observational cohort study was conducted across three tertiary hepatology centres in mainland China (Beijing, Shanghai, and Guangzhou) and enrolled participants between January 2018 and December 2022. The study was approved by the institutional review boards of all three participating centres, and written informed consent was obtained from every participant before enrolment. Reporting follows the STROBE statement for observational cohort studies.

Eligible participants were adults aged 18 years or older with a diagnosis of MASLD confirmed by liver biopsy at baseline. Individuals were excluded if they had concurrent hepatitis B or hepatitis C virus infection, reported alcohol consumption exceeding 30 g/day (men) or 20 g/day (women), had a prior or current diagnosis of any malignancy, or were pregnant at the time of enrolment. Of 2,412 participants enrolled at baseline, 425 (17.6%) were lost to follow-up before the protocol-specified second biopsy, yielding a final analytic sample of 1,987 participants. The primary exposure comprised four non-invasive fibrosis scores assessed at baseline: FIB-4, NFS, APRI, and liver stiffness measured by transient elastography (FibroScan; Echosens, Paris, France) following the manufacturer's standardised protocol. The primary outcome was fibrosis progression, defined as an increase of at least one stage on the METAVIR scale between the baseline and follow-up biopsies. Covariates specified *a priori* included age, sex, body-mass index, diabetes status, alcohol intake, and baseline fibrosis stage.

Laboratory variables were obtained using harmonised protocols across all three centres to minimise inter-site measurement variability. Liver biopsies were scored by local pathologists using the METAVIR system; to assess inter-rater reliability, a central pathologist independently re-read a random 20% sample (Cohen's $\kappa = 0.78$, indicating substantial agreement). Three potential sources of bias were identified and addressed prospectively. First, healthy-volunteer bias arising from selective refusal of the follow-up biopsy was examined in a sensitivity analysis using inverse-probability weighting (IPW), with weights derived from logistic regression on baseline clinical characteristics. Second, inter-centre misclassification of fibrosis stage was mitigated by the central re-read described above. Third, residual confounding from unmeasured dietary factors was quantified using the E-value approach.

The target sample size was determined by a pre-specified power calculation: 1,800 participants were required to achieve 80% power to detect a hazard ratio of 1.5 for fibrosis progression (two-tailed $\alpha = 0.05$), assuming a 30% three-year progression rate. An additional 612 participants were recruited to allow for the anticipated 17–20% loss to follow-up, yielding the enrolled baseline cohort of 2,412.

The primary analysis used Cox proportional-hazards regression with time-varying covariates to estimate the association between each baseline non-invasive score and the hazard of fibrosis progression. All models were adjusted for age, sex, body-mass index, diabetes status, alcohol intake, and baseline fibrosis stage. Missing covariate data were handled by multiple imputation using chained equations ($m = 10$ imputed datasets; `mice 3.16.0`). Estimates were pooled across imputed datasets using Rubin's rules. Sensitivity analyses comprised complete-case analysis and E-value calculation for unmeasured confounding. Three secondary endpoints were pre-specified; the family-wise type I error rate for these was controlled by Bonferroni correction. All tests were two-tailed with a significance threshold of $\alpha = 0.05$. Analyses were performed in `R 4.3.1` using the `survival 3.5-7` and `mice 3.16.0` packages. Median follow-up was 3.4 years (interquartile range 2.6–4.1 years).